

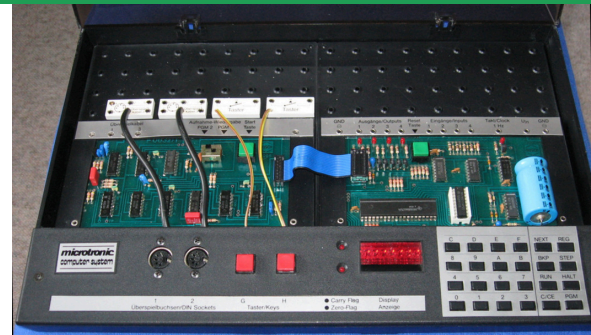
# THREE GERMAN EDUCATIONAL COMPUTERS

For a few years in the early 1980s a new category appeared in German department stores: the **educational & experimentation computer** — a complete “learn how a computer works” machine for interested laypeople, not a bare kit. Three contenders stood out, and each wins a different crown.

## Busch Microtronic

1981 · DM 379 (later 299)

THE EDUCATIONAL COMPUTER



|           |                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------|
| CPU       | TI TMS1600, 4-bit                                                                                                  |
| Clock     | 500 kHz                                                                                                            |
| Runs      | interpreted VM · 41 instructions                                                                                   |
| Specialty | powerful high-level ops — Harvard: mult, div, time, RND, ...                                                       |
| Memory    | 256 words / bank                                                                                                   |
| Speed     | ≈ 114 instr/sec                                                                                                    |
| I/O       | 4 in · 4 out (4 LEDs) · 2 input buttons (G/H) · 1 Hz, Carry & Zero LEDs · 6-digit display · hex pad · RESET button |
| Exp. bus  | — none (only a GPIO-like DIP socket)                                                                               |
| Cassette  | additional 2095 (into that socket)                                                                                 |

**Wins because:** a near-RISC, almost fully orthogonal instruction set — powerful programs in tiny memory — and manuals ideal for children and teenagers.

## Kosmos CP1

1983 · DM 198

THE EXPERIMENTATION COMPUTER



|           |                                                                               |
|-----------|-------------------------------------------------------------------------------|
| CPU       | Intel 8049, 8-bit                                                             |
| Clock     | 6 MHz                                                                         |
| Runs      | decimal interpreted VM                                                        |
| Specialty | realistic ISA — von-Neumann: indirect addressing & jumps, self-modifying code |
| Memory    | 128 → 256 cells                                                               |
| Speed     | ≈ 3,200 instr/sec                                                             |
| I/O       | many fast GPIOs · 6-digit LED · membrane                                      |
| Exp. bus  | yes — module bus (CP2/CP3/CP4/CP5)                                            |
| Cassette  | additional CP2 module                                                         |

**Wins because:** the richest, fastest I/O for real measurement & control — though a system of *squandered potential* (a few more opcodes would have worked wonders).

## Philips MasterLab

1983 · DM 449

THE REAL CPU TRAINER



|           |                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------|
| CPU       | National INS8070 (SC/MP III)                                                                                |
| Clock     | 4 MHz                                                                                                       |
| Runs      | native machine code                                                                                         |
| Specialty | true 16-bit ops — hardware MPY & DIV                                                                        |
| Memory    | 1 KB RAM                                                                                                    |
| Speed     | ≈ 147,050 instr/sec                                                                                         |
| I/O       | only SA/SB in + F1–F3 out (LEDs) · 8 output LEDs · 8-digit display · RESET + ON/OFF (the only power switch) |
| Exp. bus  | yes — but needs a custom decoder                                                                            |
| Cassette  | built-in — on board (no module)                                                                             |

**Wins because:** the only one programmed in **raw machine language** — the fastest and most “real” — but it arrived late (1983/84) and dearer.

### ⌚ SPEED — MEASURED INSTRUCTIONS PER SECOND (log scale — the MasterLab runs ~1,290× the Microtronic, ~46× the CP1)



### ⚖️ THE VERDICT — EACH WINS A DIFFERENT CATEGORY

None of the three subsumes the others. The **Microtronic** is the clear **educational computer** (near-RISC instruction set, manuals made for kids); the **CP1** is the **experimentation** champion (plentiful, fast GPIOs for measurement & control); the **MasterLab** is the only genuine **CPU trainer** (native code, interrupts, blistering speed). Scored across nineteen equally-weighted criteria, it is remarkably close — and together they are a fascinating piece of technical history that existed in this form nowhere else.

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KOSMOS CP1

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MASTERLAB



#### ► Read the full story

The complete all-in-one English write-up — 22 pages with a print-ready PDF, benchmarks, and the modern revival of all three machines — translated from the original German LOAD articles.  
[github.com/lambdamikel/german-educational-computers-of-the-80s](https://github.com/lambdamikel/german-educational-computers-of-the-80s)